3(a)(i)	amplitude = 4.9 cm	**A1**
3(a)(ii)	frequency = 2700 / 60 = 45 Hz	**A1**
3(a)(iii)	$v_0 = x_0\omega$ **and** $\omega = 2\pi f$	**C1**
	$v_0 = 4.9 \times 10^{-2} \times 2\pi \times 45$ $= 14\ \text{m s}^{-1}$	**A1**
3(a)(iv)	$v = \omega\,(x_0^2 - x^2)^{½}$ $= 2\pi \times 45 \times [(4.9 \times 10^{-2})^2 - (2.6 \times 10^{-2})^2]^{½}$	**C1**
	$= 12\ \text{m s}^{-1}$	**A1**

Question	Answer	Marks
3(b)	$F = ma$ **and** $a_0 = v_0\omega$ **or** $a_0 = x_0\omega^2$	**C1**
	$F = 0.64 \times 13.9 \times 2\pi \times 45$ **or** $0.64 \times 4.9 \times (2\pi \times 45)^2$	**C1**
	$= 2500$ N	**A1**

Question	Answer	Marks